

ENGGEN 769 - Research Methods for Engineers

Development of research methods knowledge and skills including research philosophy and design, research ethics, data collection and analysis techniques, identification of limitations, and writing up and reporting. Qualitative and quantitative research methods are addressed.

Course Details

Course Code	ENGGEN 769
Course Title	Research Methods for Engineers
Course Department	Faculty of Engineering
Course Level	Stage 7 course
Points	15
Delivery Mode	Campus Experience or Online

Course Offerings

The information below indicates when the course is normally offered. Availability may change up until semester start.

Campus	Semester Usually Offered
City	Semester One
UoA Online	Semester One
City	Semester Two

Course Overview

The course is designed as an introduction to research, specifically for those with little to no research experience, who are either about to start a research project as part of their degree or want to gain research skills in preparation for the workforce. The expectation is that by the end of the course, you will have a toolbox of skills to be able to help you on your research journey.

The course itself is divided into three parts.

- The first part is intended to give you a good sense of the context of the world of research and how to position yourself as a researcher, and your research, within this context, before moving

on to strategies for sorting information to allow for the identification of research gaps, and ultimately the important research questions that need addressing.

- The second part of the course focuses on methods, for example, how to choose an appropriate methodology and how to analyse your results so that you can draw meaningful conclusions from them.
- The third part of the course is focused on presenting your results to the world, both in written and oral formats.

Course Requisites

Restriction: must not have completed CIVIL 705, COMPSYS 700, ELECTENG 700, ENGSCI 700, MECHENG 700, SOFTENG 700, CHEMMAT 751, ENGGEN 721

Course Learning Outcomes

CLO #	Outcome	Programme Capability Link
1	Identify appropriate solutions to ethical issues in relation to research	'MEngSt - Master of Engineering Studies - Programme Capabilities Apply ethical principles and commit to professional ethics and norms of engineering practice, adhering to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WA8)
2	Critique components and forms of research outputs	'MEngSt - Master of Engineering Studies - Programme Capabilities Analyse and evaluate the impact of professional engineering work to solve complex engineering problems in societal and environmental contexts with holistic considerations for sustainable development (WA7)
3	Create a component of a scientific paper or presentation that is clear, unambiguous and informative	'MEngSt - Master of Engineering Studies - Programme Capabilities Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation, considering multiple perspectives and knowledge systems to develop solutions to complex engineering problems (WA1) Use knowledge of mathematics, natural sciences and engineering principles, and research literature to identify, formulate, analyse and solve complex engineering problems and reach substantiated conclusions (WA2)
4	Identify and apply appropriate research methods	'MEngSt - Master of Engineering Studies - Programme Capabilities Create, select, apply, and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including, measurements, modelling and prediction, to solve complex engineering problems (WA5)

Additional Information on Assessment

A passing mark is 50% or higher, according to University policy.

Late submissions will not be accepted/allowed for the exam, tests, and online quizzes. It is possible to submit late for the assignments and projects, and the late submission penalty schedule will be included in the assignment and project handouts.

Assessment Summary

#	Type	%	Classification
1	Research Literacy	35%	Individual Coursework
2	Research Design Exercise	50%	Group & Individual Coursework
3	Research dissemination	15%	Individual Coursework

Assessment Type to Learning Outcome Map

	1	2	3	4
Research Literacy		•	•	•
Research Design Exercise	•		•	•
Research dissemination		•	•	

Workload Expectations

This course is a standard 15-point course. Students are expected to spend about 10 hours per week on the course.

Every week, you can expect to spend about five hours a week working through lecture material (including two hours of lectures), one hour participating in a tutorial, and four hours working on assignments.

Delivery Mode Details

Attendance is expected at scheduled activities including tutorials to complete components of the course. Lectures will be available as recordings. Other learning activities including tutorials will not be available as recordings.

The activities for the course are scheduled as a standard weekly timetable.

Learning Resources

Taught courses use a learning and collaboration tool called Canvas to provide students with learning materials including reading lists and lecture recordings (where available). Please remember that the recording of any class on a personal device requires the permission of the instructor.

Additional Information on Learning Resources

All course material will be made available on CANVAS, the university's online learning management system. The following textbook is a useful resource for the course (available from the library as an on-line resource): Research Methods for Engineers. David V. Heiel. Cambridge University Press, 2014. ISBN 978-1-107-61019-4.

Health and Safety

Students are expected to adhere to the guidelines outlined in the Health and Safety section of the Engineering Undergraduate Handbook.

Disclaimer

Elements of this outline may be subject to change. The latest information about taught courses is made available to enrolled students in Canvas.

Students may be asked to submit assessments digitally. The University reserves the right to conduct scheduled tests and examinations online or through the use of computers or other electronic devices. Where tests or examinations are conducted online remote invigilation arrangements may be used. In exceptional circumstances changes to elements of this course may be necessary at short notice. Students enrolled in this course will be informed of any such changes and the reasons for them, as soon as possible, through Canvas.